Quantification: 701_z36c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 7 |
| interior lacuna count | 7 |
| frame-edge lacuna count | 0 |

Interior lacunae only: mean, median and sample SD (n = 7)
---------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 747.333 | 620.5 | 403.684 | px^2 |
| major_axis_length_px | lacuna shape | 43.0444 | 36.4866 | 17.4393 | px |
| minor_axis_length_px | lacuna shape | 24.5042 | 20.2991 | 7.4432 | px |
| aspect_ratio | lacuna shape | 1.8392 | 1.5104 | 0.8934 | unitless |
| eccentricity | lacuna shape | 0.7622 | 0.7481 | 0.1225 | unitless |
| solidity | lacuna shape | 0.8156 | 0.8224 | 0.0898 | unitless |
| roots_count | headline | 8.1429 | 7 | 4.1404 | count |
| ring_length_r30_px | headline | 409.429 | 366 | 158.137 | px |
| ring_length_r60_px | reported | 1129.43 | 1091 | 359.696 | px |
| owned_length_px | ownership-dependent | 3076.75 | 1854.51 | 2928.18 | px |
| edge_count | ownership-dependent | 230.286 | 129 | 224.427 | count |
| mean_edge_length_px | ownership-dependent | 13.7513 | 13.3037 | 1.3123 | px |
| perimeter_px | normalised | 136.282 | 105.332 | 59.1302 | px |
| ring_area_r30_px2 | normalised | 6094.14 | 5383 | 1193.79 | px^2 |
| ring_area_r60_px2 | normalised | 17768.4 | 16362 | 2341.76 | px^2 |
| in_frame_fraction_r30 | normalised | 1 | 1 | 0 | unitless |
| in_frame_fraction_r60 | normalised | 1 | 1 | 0 | unitless |
| ring_density_r30 | normalised | 0.0665947 | 0.064681 | 0.0185438 | px^-1 |
| ring_density_r60 | normalised | 0.0630826 | 0.0590751 | 0.0163286 | px^-1 |
| roots_per_100px_perimeter | normalised | 6.4764 | 7.1514 | 2.8938 | per 100 px |
| ring_attached_length_r30_px | network v2 | 343.857 | 282 | 167.613 | px |
| ring_attached_length_r60_px | network v2 | 785.857 | 542 | 521.394 | px |
| ring_length_w_r30_px | network v2 | 466.815 | 408.291 | 182.895 | px |
| ring_length_w_r60_px | network v2 | 1276.87 | 1215.15 | 416.559 | px |
| sholl_crossings_r10 | network v2 | 8.1429 | 7 | 2.7946 | count |
| sholl_crossings_r20 | network v2 | 9.8571 | 9 | 4.0999 | count |
| sholl_crossings_r30 | network v2 | 13.8571 | 17 | 4.5981 | count |
| loop_count | loops, report only | 27.4286 | 8 | 39.8617 | count |
| loop_edge_share | loops, report only | 0.2935 | 0.2713 | 0.2135 | share |
| ring_width_mean_r30_px | width | 6.3232 | 5.8991 | 1.0478 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.0438e+06 | px^2 |
| total_skeleton_length_px | 50926 | px |
| canalicular_length_density_per_px | 0.048789 | px^-1 |
| junction_count | 1527 | count |
| junction_density_per_px2 | 0.00146292 | px^-2 |
| lacuna_count | 7 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 6.71e-06 | px^-2 |
| skeleton_component_count | 921 | count |
| mean_component_length_px | 55.2942 | px |
| median_component_length_px | 12 | px |
| field_length_density_w_per_px | 0.0547478 | px^-1 |
| field_density_without_flagged_per_px | 0.0548852 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 64 | px |
| unattached_component_count | 5 | count |
| canalicular_length_density_attached_per_px | 0.048789 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0488503 | px^-1 |
| loop_count | 245 | count |
| loop_edge_share | 0.244 | share |
| edge_count | 3934 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 6.0111 | px |
| width_median_px | 6 | px |
| width_p10_px | 2 | px |
| width_p90_px | 10 | px |
| width_pixels_used | 33988 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 804 | 41.5213 | 33.4623 | 1.2408 | 0.592 | 0.6949 | -0.5169 | 176.293 | 670.124 | False |
| 2 | 437 | 31.4519 | 18.3155 | 1.7172 | 0.8129 | 0.8918 | 1.4887 | 285.545 | 929.915 | False |
| 3 | 406 | 28.9527 | 20.3619 | 1.4219 | 0.7109 | 0.7823 | -0.2801 | 325.505 | 634.365 | False |
| 4 | 1410 | 54.1218 | 34.645 | 1.5622 | 0.7683 | 0.921 | -1.5551 | 413.697 | 828.063 | False |
| 5 | 994 | 72.7028 | 20.0045 | 3.6343 | 0.9614 | 0.7412 | 1.4325 | 768.803 | 413.569 | False |
| 6 | 433 | 29.5156 | 20.2362 | 1.4586 | 0.728 | 0.8625 | 0.4576 | 943.721 | 446.413 | False |
| 7 | | | | | | | | 593.976 | 873.082 | False |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 8 | 443 | 1213 | 3936.78 | 314 | 12.5375 | 5.8682 |
| 2 | 7 | 295 | 903 | 1330.54 | 91 | 14.6214 | 6.9536 |
| 3 | 4 | 230 | 655 | 98.0833 | 8 | 12.2604 | 5.8913 |
| 4 | 6 | 366 | 1091 | 782.985 | 49 | 15.9793 | 4.7476 |
| 5 | 17 | 714 | 1722 | 5219.57 | 396 | 13.1807 | 6.9086 |
| 6 | 7 | 476 | 1424 | 8314.79 | 625 | 13.3037 | 7.9939 |
| 7 | 8 | 342 | 898 | 1854.51 | 129 | 14.376 | 5.8991 |

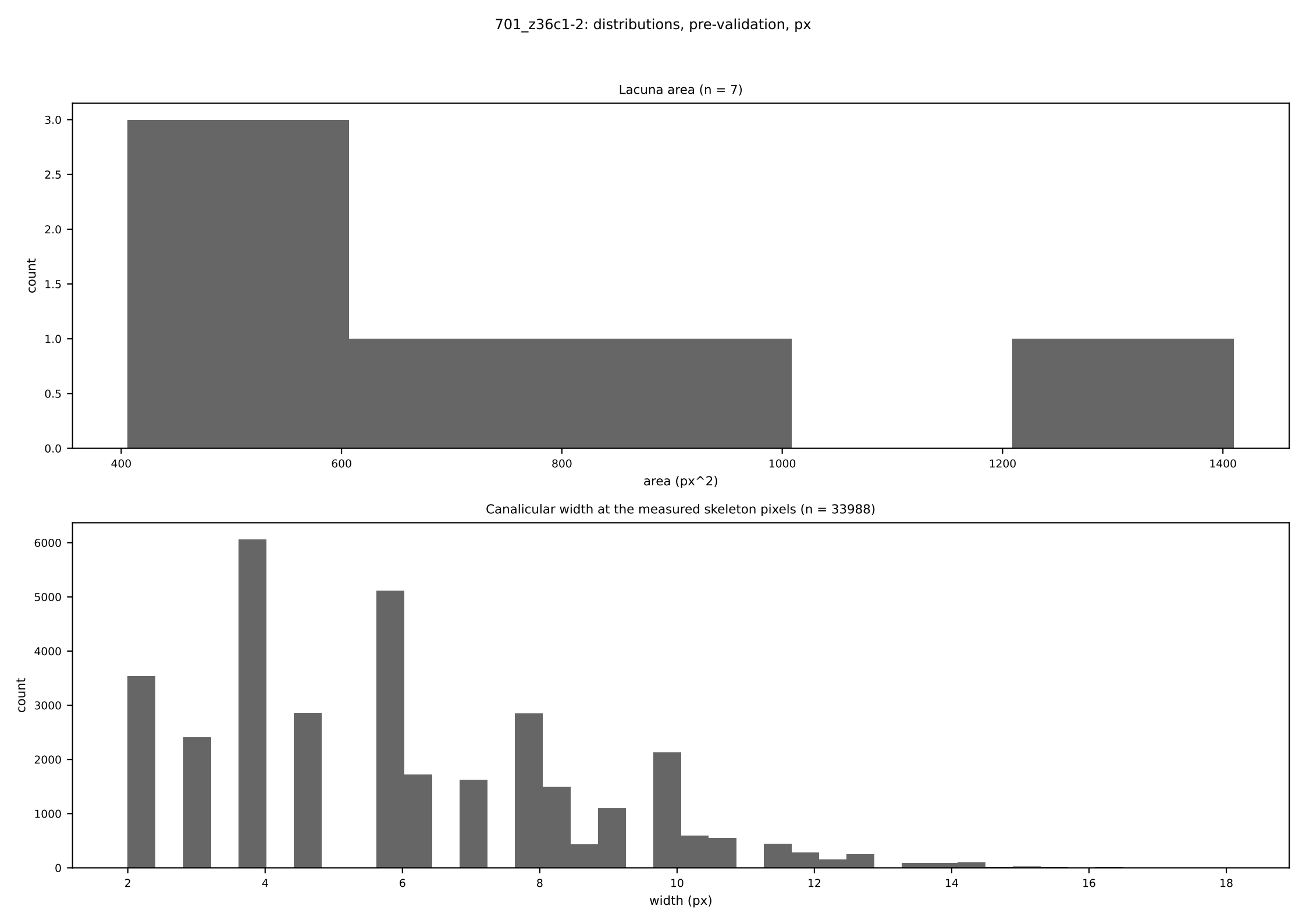
Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 227.635 | 6849 | 19116 | 1 | 1 | 0.064681 | 0.0634547 | 3.5144 | |
| 2 | 94.468 | 5255 | 16151 | 1 | 1 | 0.056137 | 0.0559098 | 7.4099 | |
| 3 | 105.332 | 5383 | 16362 | 1 | 1 | 0.0427271 | 0.0400318 | 3.7975 | |
| 4 | 158.267 | 7198 | 20036 | 1 | 1 | 0.0508475 | 0.054452 | 3.7911 | |
| 5 | 198.309 | 7886 | 21287 | 1 | 1 | 0.0905402 | 0.0808944 | 8.5725 | |
| 6 | 97.8823 | 5305 | 16226 | 1 | 1 | 0.0897267 | 0.0877604 | 7.1514 | |
| 7 | 72.0833 | 4783 | 15201 | 1 | 1 | 0.0715032 | 0.0590751 | 11.0983 | |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 372 | 884 | 508.617 | 1375.79 | 11 | 9 | 18 |
| 2 | 278 | 514 | 344.463 | 1017.09 | 6 | 9 | 9 |
| 3 | 165 | 265 | 261.794 | 735.926 | 5 | 7 | 8 |
| 4 | 194 | 320 | 408.291 | 1215.15 | 10 | 6 | 17 |
| 5 | 640 | 1571 | 821.955 | 1973.5 | 12 | 17 | 18 |
| 6 | 476 | 1405 | 539.848 | 1610.23 | 7 | 14 | 17 |
| 7 | 282 | 542 | 382.735 | 1010.41 | 6 | 7 | 10 |


701_z36c1-2: distributions, pre-validation, px
Lacuna area (n = 7)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 33988)
count
width (px)